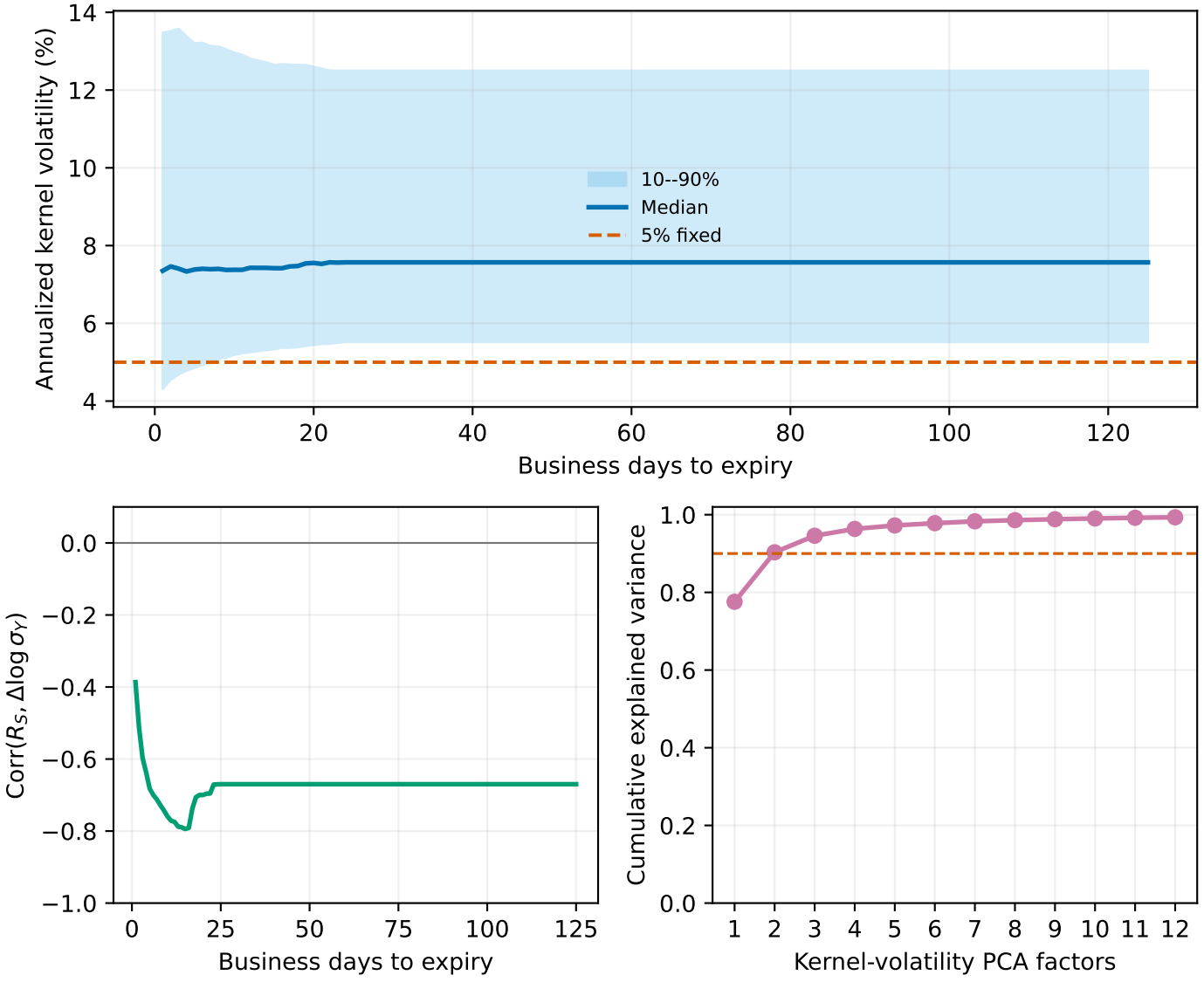
Annualized kernel volatility (%)
14
12
10
8
6
4
10--90%
Median
5% fixed
0
20
40
60
80
100
120
Business days to expiry
Corr(R_S, $\Delta \log \sigma_Y$)
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
0
25
50
75
100
125
Business days to expiry
Cumulative explained variance
1.0
0.8
0.6
0.4
0.2
0.0
1
2
3
4
5
6
7
8
9
10
11
12
Kernel-volatility PCA factors